

Polynomial Regression Assignment

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1 Introduction

This assignment uses polynomial regression to predict a continuous target for two personalized datasets. VAR1 is the Power Plant Steam Turbine Optimization problem, and VAR2 is the Subterranean Thermal Reservoir Mapping problem. The target column is y in both cases. The test targets are hidden, so I used the test files only to generate predictions after choosing the models with training data.

2 Data

VAR1 has six turbine inputs: steam valve adjustment, condenser coolant flow, reinjection pump pressure, blade pitch, exhaust valve rate, and steam inlet pressure (x_1 – x_6). Its target is the Net Power Score. VAR2 has three spatial inputs: east-west offset, north-south offset, and vertical depth (x_1 – x_3); its target is the Thermal Anomaly Score.

Each variant has 1,000 training rows and 1,000 test rows. All feature values are between -1 and 1 . There were no missing values, and the train and test feature columns matched. The duplicate counts below describe exact duplicate rows in each file; test rows were kept in their original order. The target summaries give a quick sense of the scale and spread of the training responses. They were not used to transform the targets.

Table 1: Summary of the supplied datasets.

Variant	Train / test rows	Inputs	Target range	Target mean (SD)	Duplicates (train / test)
VAR1	1,000 / 1,000	6	$-7.873, 11.900$	0.853 (3.106)	0 / 8
VAR2	1,000 / 1,000	3	$-29.609, 39.518$	2.182 (6.605)	0 / 10

3 Method

I expanded the input columns into polynomial terms, including interactions, then compared ordinary least squares (OLS), Ridge, Lasso, and ElasticNet. For the regularized models, I tried both scaled and unscaled polynomial features. The candidate degree ranges were 2–7 for VAR1 and 6–14 for VAR2.

The polynomial expansion included interaction terms and did not add a separate bias column. Scaling was applied after expansion for the scaled candidate pipelines. This is useful for Lasso because its penalty acts on the coefficients; scaling puts polynomial columns on a more comparable scale before that penalty is applied. The final VAR2 Ridge pipeline did not use scaling because the unscaled version was selected in validation.

Model selection used nested cross-validation on the training data. Five outer folds estimated performance, and three inner folds selected the regularization settings. The folds were shuffled with random state 7. I compared mean squared error (MSE) and R^2 , using MSE as the main criterion. For candidates within one standard error of the lowest mean MSE, I favoured the simpler polynomial basis. The test data was not used for model selection or hyperparameter tuning.

4 Results and Model Selection

Table 2 summarizes the chosen configurations and compares their outer-fold results with the original OLS baselines. The reported CV values are training-only estimates, not hidden-test scores.

Table 2: Selected models and OLS baselines from outer-fold cross-validation.

Variant	Chosen model	Degree	CV MSE (mean $\pm$ SD)	CV R^2 mean	OLS baseline (degree, MSE, R^2)
VAR1	Scaled Lasso	5	0.3491 ± 0.0384	0.9631	4, 0.7717, 0.9185
VAR2	Ridge	10	0.2329 ± 0.0134	0.9943	8, 0.2556, 0.9937

For VAR1, the lowest mean CV MSE was 0.3347 for unscaled degree-5 Lasso. Scaled degree-5 Lasso had MSE 0.3491, a difference of 0.0144, which is within the one-standard-error allowance of 0.0202. It also used fewer nonzero coefficients on average across the outer folds (101.6 versus 119.6), so I chose the scaled version. The final alpha was retuned using five-fold CV on all training rows and was 0.006.

For VAR2, degree-11 Ridge had the lowest mean CV MSE at 0.2327. Degree-10 Ridge scored 0.2329, only 0.0002 higher, within the one-standard-error allowance of 0.0069. Degree 10 uses a smaller basis (285 rather than 363 terms on average), so I selected it. The final alpha was 0.1.

The alpha values in the final pipelines were tuned again using five shuffled folds on all training rows (random state 42), after choosing the model family, degree, and scaling option. That gave final alpha 0.006 for VAR1 and 0.1 for VAR2. Test features and hidden test targets were not part of this step.

Table 3: Close candidates considered under the one-standard-error rule.

Variant	Candidate	CV MSE (mean $\pm$ SD)	Mean nonzero terms	Within one SE?
VAR1	Degree-5 Lasso, unscaled	0.3347 ± 0.0452	119.6	Lowest MSE
VAR1	Degree-5 Lasso, scaled	0.3491 ± 0.0384	101.6	Yes; selected
VAR2	Degree-11 Ridge	0.2327 ± 0.0153	363.0	Lowest MSE
VAR2	Degree-10 Ridge	0.2329 ± 0.0134	285.0	Yes; selected

The comparison with OLS favoured the selected regularized models in these folds. For VAR1, the selected CV MSE was 0.3491 versus 0.7717 for degree-4 OLS. For VAR2, it was 0.2329 versus 0.2556 for degree-8 OLS. No test MSE or test R^2 is reported because the test labels are hidden.

In terms of mean CV MSE, these correspond to reductions of about 54.8% for VAR1 and 8.9% for VAR2 relative to the stated OLS baselines. This is only a comparison of the validation estimates; it does not predict the hidden-test score.

5 VAR1 Figures

The figures show cross-validation error by degree, out-of-fold predictions, residuals, and the number of nonzero Lasso coefficients. The prediction and residual plots use out-of-fold predictions, so each plotted row was held out while its fold model was fitted.

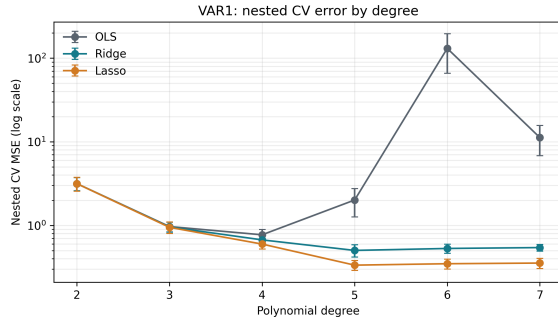


Figure 1: VAR1 CV MSE by polynomial degree.

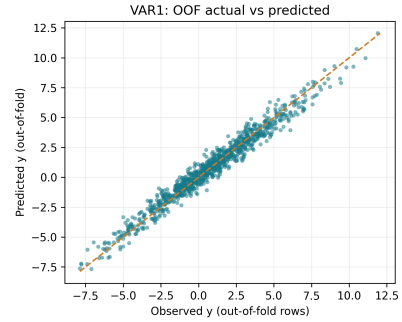


Figure 2: VAR1 out-of-fold actual values and predictions.

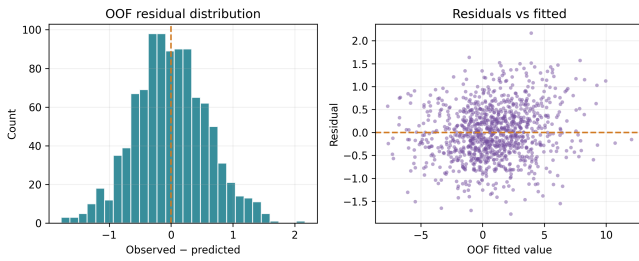


Figure 3: VAR1 out-of-fold residual diagnostics.

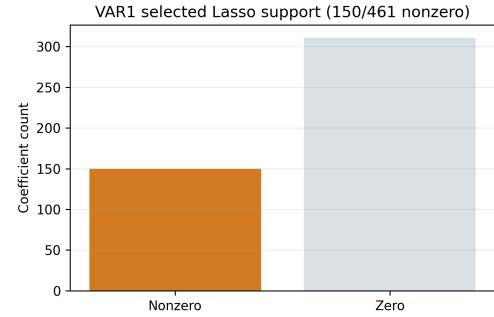


Figure 4: Nonzero coefficients in the final VAR1 Lasso fit.

The actual-versus-predicted points mostly follow the diagonal, and the residuals cluster around zero. The final full-data Lasso fit retained 150 nonzero polynomial coefficients.

6 VAR2 Figures and Stability Check

The first plots show the degree comparison and out-of-fold predictions. The residual plot gives a second view of the fold predictions.

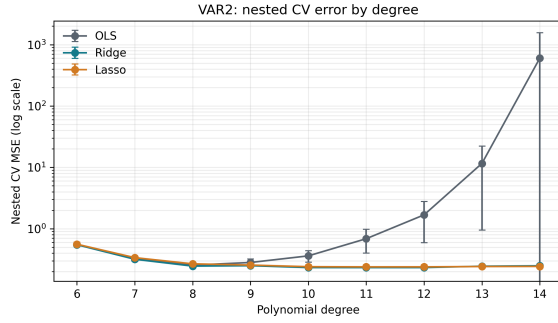


Figure 5: VAR2 CV MSE by polynomial degree.

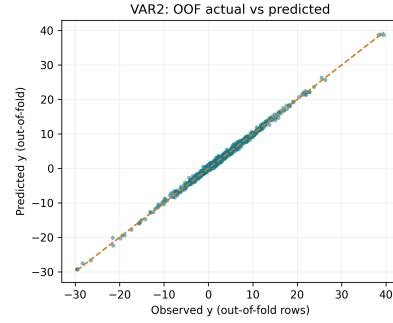


Figure 6: VAR2 out-of-fold actual values and predictions.

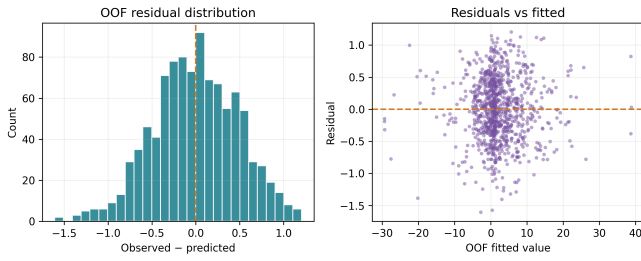


Figure 7: VAR2 out-of-fold residual diagnostics.

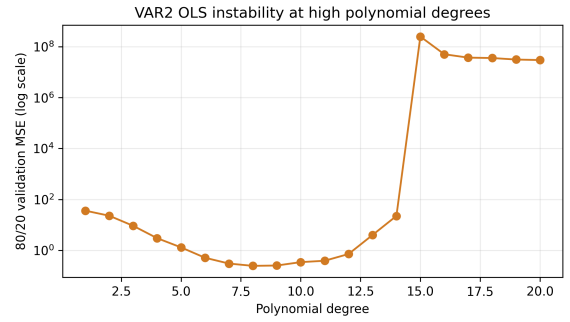


Figure 8: VAR2 OLS validation MSE by polynomial degree.

I also checked high-degree OLS on a separate 80/20 split of the VAR2 training data (random state 42). At degree 15, the design matrix rank was 793 out of 815 columns and the smallest-to-largest singular-value ratio was 6.8×10^{-17} . At degree 20, the rank was 793 out of 1,770 columns, the ratio was 7.6×10^{-17} , and validation MSE was about 29.9 million. These results show why the maximum degree was not automatically chosen.

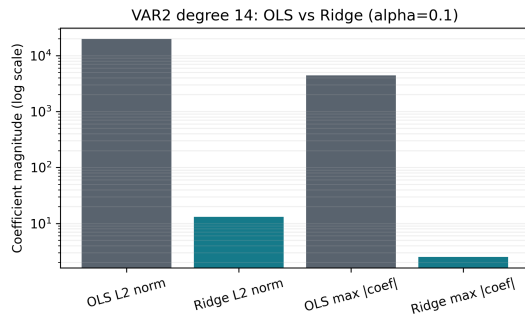


Figure 9: Coefficient sizes for degree-14 VAR2 OLS and Ridge.

7 Final Models and Predictions

After selecting the configurations, I refit each model on all 1,000 training rows and predicted all 1,000 matching test rows in their original order. The final VAR1 pipeline used degree-5 polynomial features, standard scaling, and Lasso with alpha 0.006. VAR2 used degree-10 polynomial features and Ridge with alpha 0.1, without scaling.

The selected polynomial expansions contain 461 terms for VAR1 and 285 terms for VAR2, excluding the intercept. In the final VAR1 Lasso fit, 150 coefficients were nonzero; Ridge retains all fitted terms but shrinks their coefficient values.

The prediction files are `BT2024167_pred_var1.csv` and `BT2024167_pred_var2.csv`. Each contains one numeric y column and no index column. The ranges below describe the predictions only; they are not accuracy measures.

Table 4: Summary of generated test predictions.

Variant	Minimum	Maximum	Mean	Standard deviation
VAR1	-9.765	15.182	1.282	4.215
VAR2	-29.394	38.841	2.053	6.784

The code, data, figures, and saved cross-validation results are in the [project repository](#). The training search can be run with `python train_final.py`; the selected models can be refit for inference with `python inference.py --variant var1` and `python inference.py --variant var2`.

8 Conclusion

I selected degree-5 scaled Lasso for VAR1 and degree-10 Ridge for VAR2 using training-only nested cross-validation. The one-standard-error rule supported the smaller degree-10 basis for VAR2 even though degree 11 had a slightly lower mean CV MSE. Both models were refit on all training rows before generating predictions. Their final hidden-test performance is unknown.